

Bocconi

MEASURING PERFORMANCE 30060 - MANAGEMENT

Walsh Book

Department of Management & Technology



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AGENDA

1. FINANCIAL RATIOS
2. PROFITABILITY RATIOS
3. LIQUIDITY RATIOS
4. FINANCIAL STRENGTH RATIOS
5. FINANCIAL LEVERAGE



1.0

FINANCIAL RATIOS



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P&L

REVENUES	
-	OPERATING COSTS
=	EBITDA
-	DEPRECIATIONS
-	AMORTIZATIONS
=	EBIT
-	INTERESTS
=	EBT
-	TAXES
=	NET INCOME
-	DIVIDENDS
=	RETAINED EARNINGS

↔
Ratios

Balance sheet

Assets	Liabilities
Fixed assets Net fixed assets Investments Intangibles	Owners' funds Issued common stocks Capital reserves Revenue reserves
Current assets Inventories Accounts receivable Cash and cash equivalents Miscellaneous	Long-term loans Bank loans Bonds issued
	Current liabilities Short term loans Accounts payable Miscellaneous

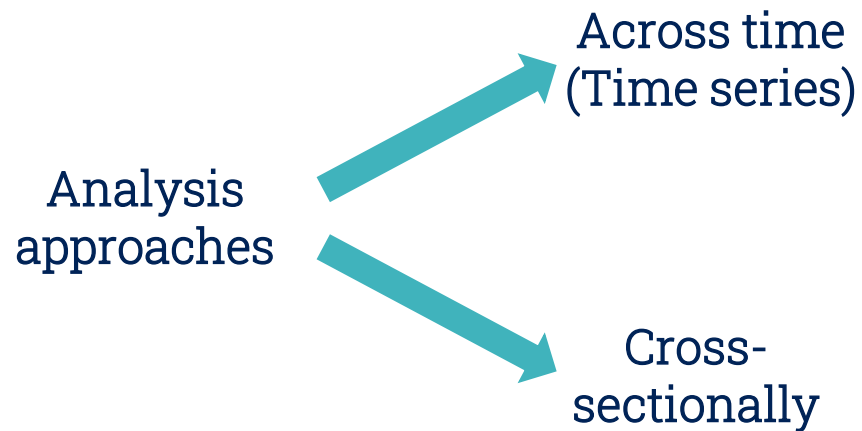


FIRM *ALPHA* 2020 ROTA = 7,4%
How can we interpret this result?

GOOD

BAD





ROTA 2011	ROTA 2012	ROTA 2013
1.3%	2.6%	7.4%

FIRM	ROTA 2013
ALPHA	7.4%
BETA	1.4%
GAMMA	2.3%
DELTA	3.3%



Financial ratios

PROFITABILITY RATIOS

measure the capacity to preserve the income equilibrium in the future

FINANCIAL STRENGTH RATIOS

measure the financial stability

LIQUIDITY RATIOS

measure solvency, i.e. the capacity for the company to pay bills as they arrive



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PROFITABILITY RATIOS



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Profitability ratios

TIME SERIES COMPARISONS

Higher profitability ratio of a company compared to its previous period indicates that the company is doing better than before, meaning that the choices taken are improving performances

CROSS-SECTIONAL COMPARISONS

If a company is having a higher profitability ratio compared to its competitors, it can be inferred that the company is doing better than competitors, meaning that choices taken are more effective than competitors' ones



Comparisons need be meaningful!

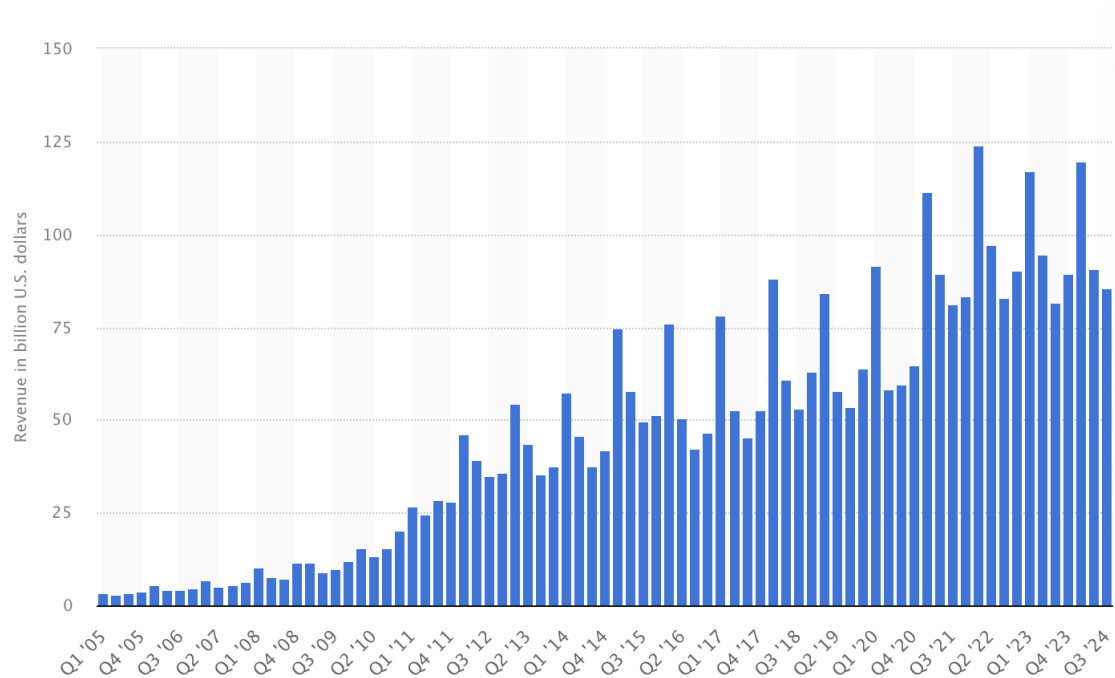
Example

Retail firms have high revenue during the Christmas season:

- Comparing the profit margin of the 4th quarter with the 1st quarter do not give a clear picture of the profitability
- To judge the profitability appropriately, the profit margin of the 4th quarter should be compared with the profit margin of the 4th quarter of the previous year



Apple's global revenue from 1st quarter 2005 to 3rd quarter 2024 (billion USD)



Return on total assets (ROTA)

$$\frac{\text{EBIT}}{\text{Total assets}}$$

EBIT

Is the amount remaining when total operating costs are deducted from total revenues

Total operating cost includes direct factory cost, administration, overheads, and depreciation and amortization

Total Assets

Resources controlled by the firm as a result of past events and from which future economic benefits are expected to flow to the firm

Current Assets + Fixed Assets



Interpreting ROTA

- Rate of return earned by the total assets
- This ratio measures how well management uses all the assets to generate an opening surplus
 - Key tool in directing management's day-to-day activities.
 - It provides a benchmark against which all operations can be measured



Problems with ROTA

- Firms deal with several products, not just one:
hard to **disentangle specific contributions** from average components
- It differs widely **across industry**
- **Depends** heavily on **asset valuations** in the Balance Sheet:
age of plants, depreciation etc. play an important role



Decomposing ROTA

However, as a single figure it simply provides a target. To be useful in decision-making, **it must first be broken down into its component parts**

$$\frac{\text{EBIT}}{\text{Total assets}} = \frac{\text{EBIT}}{\text{Total sales}} \times \frac{\text{Total sales}}{\text{Total assets}}$$

ROTA ROS Asset turnover
Return on total assets Return on sales



ROTA: interpretation

- ROTA tells you how much money is generated from each dollar invested into the firm
- It shows you the relationship between its resources (assets) and its income, providing a comparison term to evaluate if an organization is using its assets more or less effectively than in the past
- Comparisons should be made with companies **in the same industry**. When ROTA is higher than industry average, a company is said to have a **competitive advantage**.

	EBIT	Total Assets	ROTA
Apple	\$71.230 B	\$290.479 B	24.5%
Microsoft	\$19.860 B	\$193.694 B	10.2%
HP	\$5.471 B	\$106.862 B	5.1%



Return on sales (ROS)

$$\frac{\text{EBIT}}{\text{Total sales}}$$

- Also named Net profit sales percent
- Identifies profit as a percentage of sales (how much profit is being produced per dollar of sales)
- Compare a company's ROS over time to look for trends
 - An increasing ROS indicates the company is growing more efficient, while a decreasing ROS could signal troubles
- Compare it to other companies in the industry



Asset turnover

$$\frac{\text{Total sales}}{\text{Total assets}}$$

- Not a percentage relationship → How many times sales exceed total assets
- The ability of a company to generate revenue from its assets
- It is less often emphasized in the assessment of company performance
- Its contribution to ROTA is just as powerful and important as the ROS



- Companies with low-profit margins may have high asset turnover
- Grocery stores have high/low asset turnover? Why?

Asset turnover: interpretation

- Useless to compare Walmart and AT&T because they are in different sectors.
- In contrast, comparing AT&T and Verizon is useful and tells you that Verizon was more efficient at deploying its assets in generating revenues in 2016.
- However, remember to review for trends: AT&T's asset turn drastically improved since 2006

Walmart - retail

Tot.revenues (TR)
\$476 billion
Total assets (TA)
\$204 billion
Asset turn =
 $(TR/TA) =$
2.36

AT&T - telecom.

Tot.revenues
\$132 billion
Total assets
\$287 billion
Asset turn =
 $(TR/TA) =$
0.46

But asset turn in
2006 was 0.3

Verizon - telecom.

Tot.revenues
\$127 billion
Total assets
\$253 billion
Asset turn =
 $(TR / TA) =$
0.5



Exercise

Decompose ROTA into ROS and Asset Turnover. Interpret the two values obtained.

Balance Sheet		P&L	
Owners' funds	50	Revenues	100
Fixed assets	150	Operating costs	70
Inventories	50	Depreciation	0
Long-term debt	135	Interest	5
Accounts receivable	5	Tax	5
Cash	5	Dividends	10
Bank loans	20		
Bonds issued	5		



Exercise – solution (1/2)

Decompose ROTA into ROS and Asset Turnover. Interpret the two values obtained.

Balance Sheet		P&L	
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Cash	5	Dividends	10
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Bonds issued	5		

**TOT
ASSETS
= 210**

**EBIT
= 30**

$$ROA = \frac{EBIT}{TOT\ ASSETS} = \frac{30}{210} = 14.28\%$$



Exercise – solution (2/2)

Decompose ROTA into ROS and Asset Turnover. Interpret the two values obtained.

$$ROA = \frac{EBIT}{TOT\ ASSETS} = \frac{30}{210} = \text{€ } 14.28\%$$

$$ROS = \frac{EBIT}{TOT\ SALES} = \frac{30}{100} = 30\%$$

$$ASSET\ TURNOVER = \frac{TOT\ SALES}{TOT\ ASSETS} = \frac{100}{210} = 0.476$$

$$\begin{aligned} ROA &= ROS * Asset\ Turnover \\ &= 30\% * 0.476 \\ &= \text{€ } 14.28\% \end{aligned}$$



Return on equity (ROE)

$$\frac{\text{Net income}}{\text{Owners' funds}}$$

- Measures the absolute return delivered to shareholders
- A good ROE also makes it easy to attract new funds and enables the company to grow



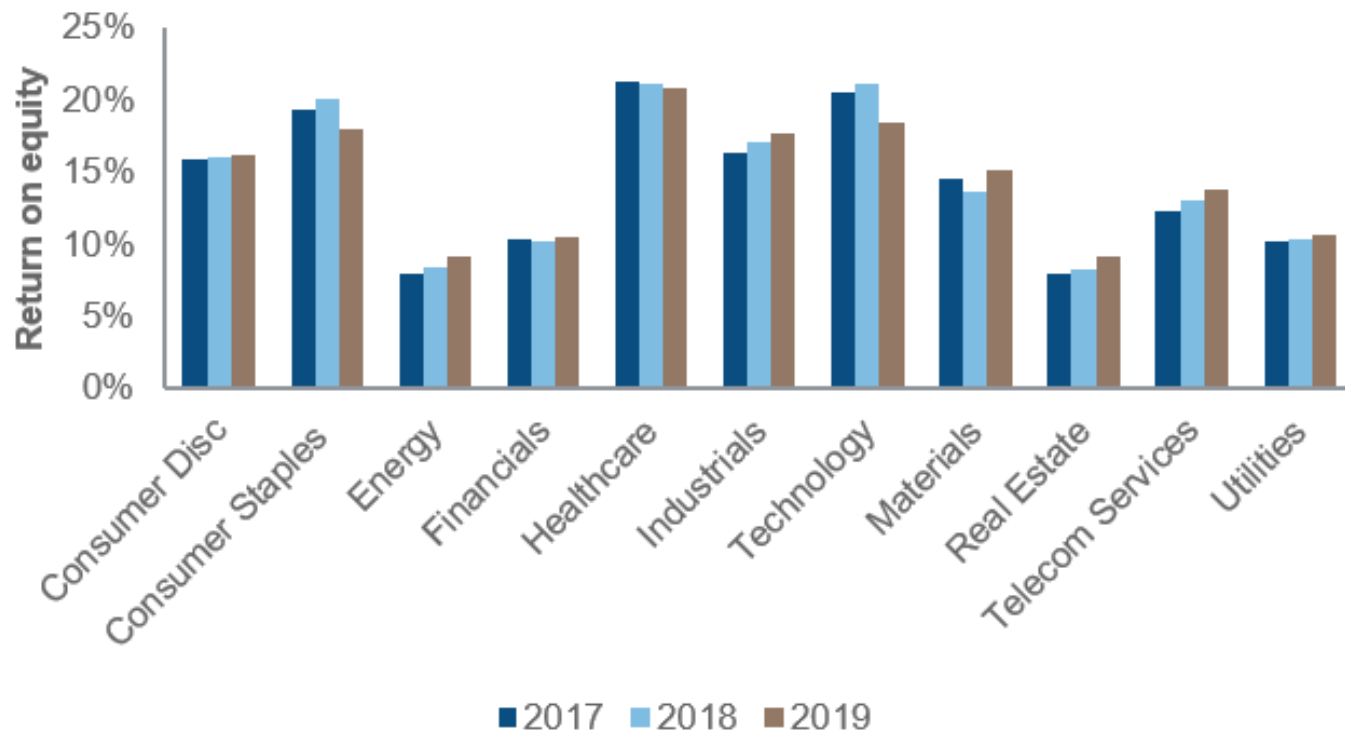
ROE: interpretation

- It should be high enough to compensate shareholders for the specific **risk** of investing in the company (higher than risk-free financial instruments).
- Riskier companies should earn higher ROE to convince shareholders to invest.

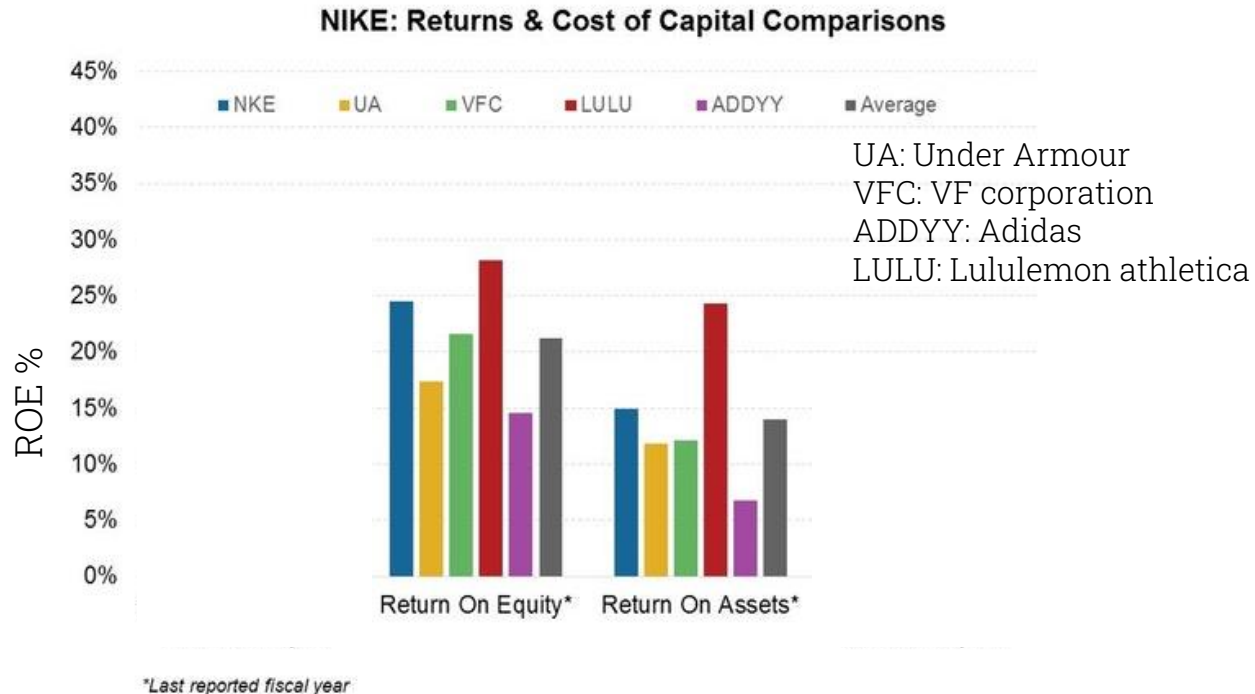
	ROTA	Net income	Owners' funds	ROE
Apple	24.5%	\$53.594 B	\$119.355 B	44.9%
Microsoft	10.2%	\$16.798 B	\$71.997 B	23.3%
HP	5.1%	\$4.554 B	\$27.768 B	16.4%



ROE across industries



ROE within an industry Apparel/footwear consumer



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LIQUIDITY RATIOS



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Liquidity

- Short/Long-term solvency, which is the capacity for the company to pay all current liabilities
- Short-term liabilities are an important way of funding firm's operations (e.g. payables)
- **Profitability and liquidity are not the same concepts** (It is possible to be profitable, having liquidity problems):
 - Not all the payments are made at the same time (e.g. Delayed Payments)
 - Not all inflows and outflows of cash are costs and revenues (e.g. Debt Reimbursement)
 - Some costs are spread over many years (e.g. Depreciation)

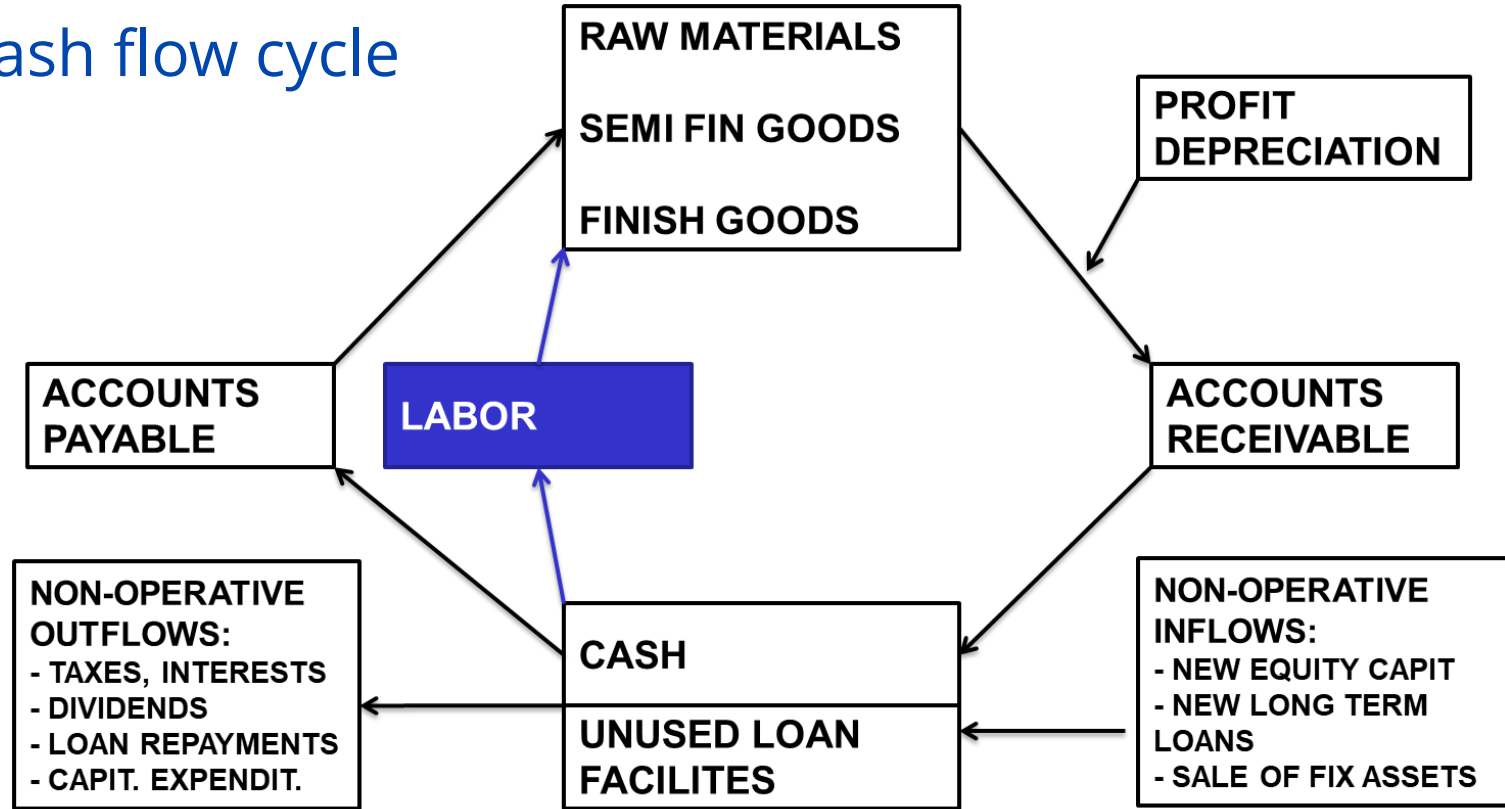


Cash flow cycle

- Cash Flow Cycle describes how the cash flows in and out of business
- Cash is a continuous circulation through the arteries of the business carrying value to the organs
- If this flow is stopped, or even seriously reduced for a time, then serious consequences result



Cash flow cycle



Current ratio

$$\frac{\text{Current assets}}{\text{Current liabilities}}$$

- A ratio greater than 1 tells us that the firm has enough near-cash assets to cover payments due in the immediate future
 - However, as all indexes, the threshold value differs drastically depending on industry etc.
 - <https://www.readyratios.com/sec/ratio/current-ratio/>
- It doesn't distinguish between different types of current assets (some are more liquid than others)



Quick ratio

$$\frac{\text{Current assets} - \text{Inventories}}{\text{Current liabilities}}$$

- We exclude inventories because its may be difficult sometimes to turn them quickly into cash
 - The numerator includes only current assets that can become cash soon (e.g. cash holdings, marketable securities)
- A value of 1 is very high, Around 0.8 is a good figure
- Widely used by lending institutions to assess companies' liquidity levels



Working capital to sales ratio

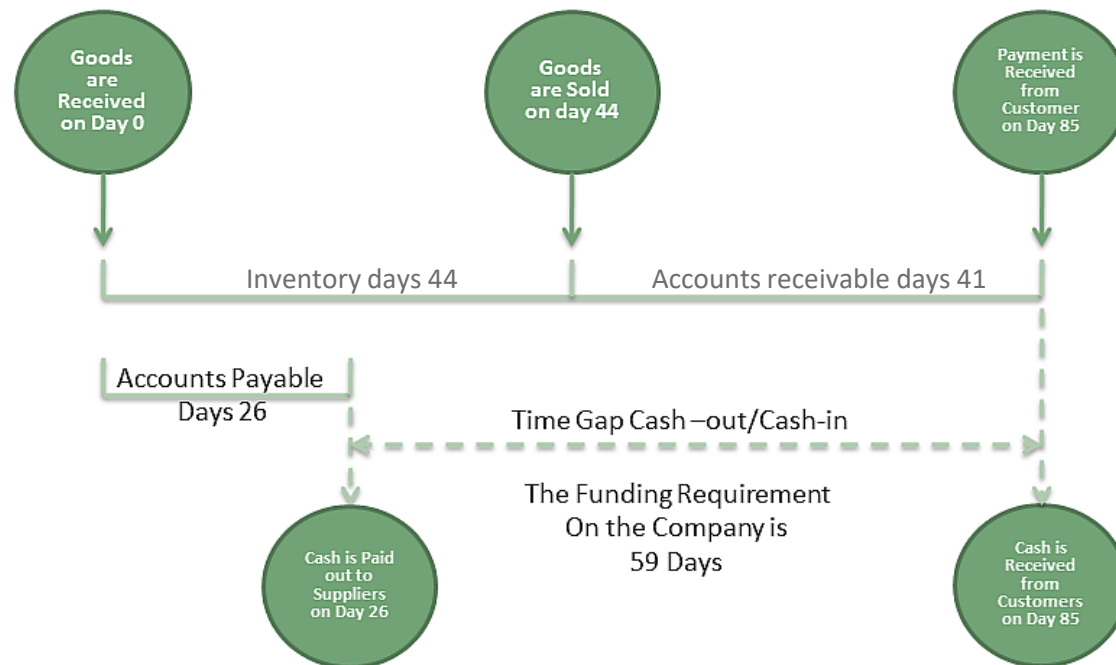
$$\frac{\text{Current assets} - \text{Current liabilities}}{\text{Sales}} \times 100$$

- Includes items from both B/S and income statement
- It measures the relationship between the funds used to finance a company's operations and the revenues a company generates
 - It says how efficiently a company is using its working capital to support sales and growth
- Values between 10% and 20% are considered solid
- Not necessarily co-move with current and quick ratio
- Remember that the other formula to compute working capital is $OF + LTD - FA$



Working capital days

Funding requirement ("Cash conversion cycle") = (Inventory days + Accounts receivable days) – Accounts payable days. **The shorter the better!**



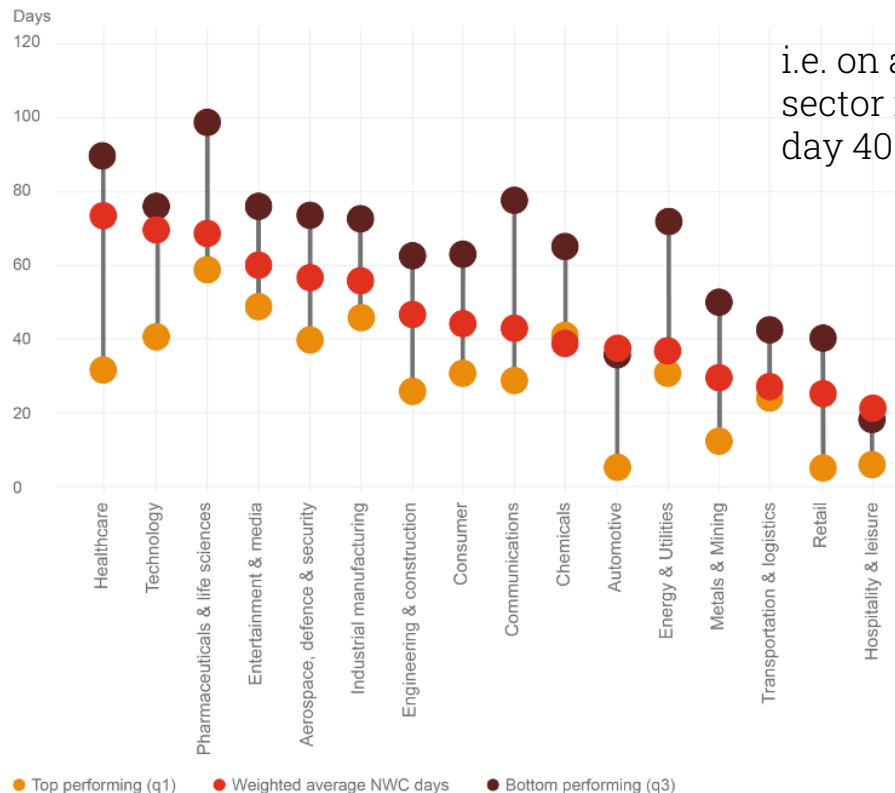
Cash conversion cycle

- Short time (effective management of inventory and credit sales)
 - The company needs to finance account receivables and inventory for a short period of time

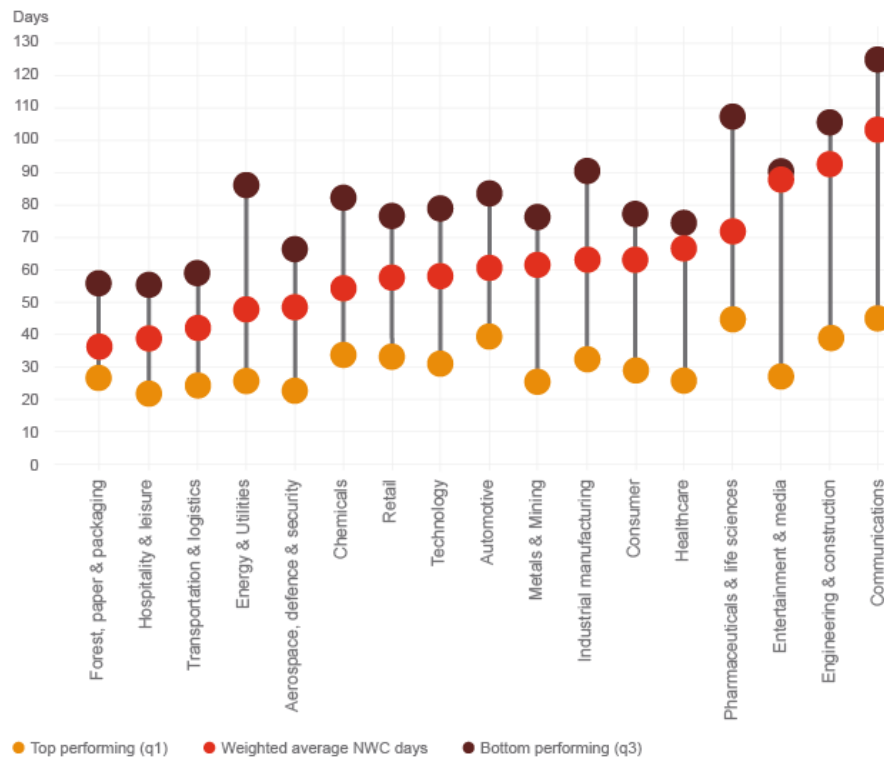
- Long time
 - Longer to sell company's products
 - Longer to receive payments from customers
 - Too quick payment of bills



Days sales outstanding sector spread in 2015



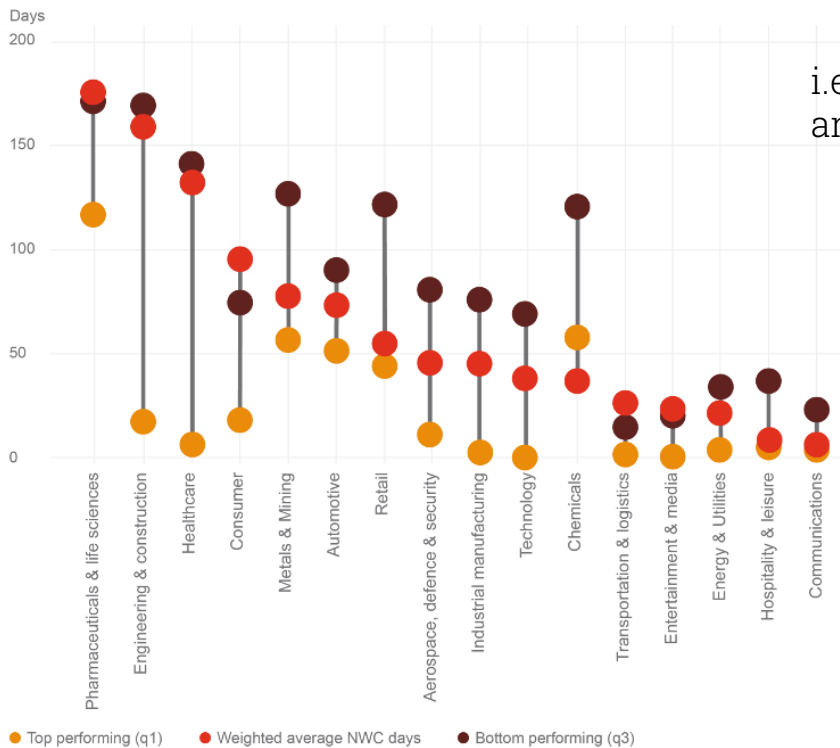
Days payable outstanding sector spread in 2015



i.e. on average the automotive sector pays around day 60



Days inventories on hand sector spread in 2015



i.e. on average the automotive has around 70 days of inventory



Cash conversion cycle: Example

AUTOMOTIVE SECTOR

Inventory days	70 days +
Receivable days	40 days -
Payable days	60 days =
Cash conversion cycle	50 days

AMAZON

Inventory days	29 days +
Receivable days	11 days -
Payable days	54 days =
Cash conversion cycle	-14 days



Working capital days

$$\text{Debtor days} = \frac{\text{Accounts receivable}}{\text{Sales}} \times 365$$

Or “collection period” (“Accounts receivable days”). **How many days on average it takes to get paid by customers** (the shorter the better)

$$\text{Inventory days} = \frac{\text{Inventories}}{\text{Sales}} \times 365$$

How many days on average an item stays in the inventory (the shorter the better)

$$\text{Creditor days} = \frac{\text{Accounts payable}}{\text{Sales}} \times 365$$

Or “Accounts payable days”. **How many days on average it takes to pay suppliers** (the longer the better)



Exercise

Compute and interpret (1) current ratio; (2) quick ratio

B/S		Income statement	
Owners' funds	40	Revenues	100
Fixed assets	160	Interest	2
Inventories	50	Tax	3
Long-term debt	135	Depreciation	10
Accounts receivable	5	Dividends	2
Cash holdings	5	Operating costs	75
Short-term debt	20		
Accounts payable	25		



Exercise – solution (1/3)

Compute and interpret (1) current ratio; (2) quick ratio, and (3) working capital to sales ratio

B/S		Income statement	
Owners' funds	40	Revenues	100
Fixed assets	160	Operating costs	75
Inventories	50	Depreciation	10
Long-term debt	135	Interest	2
Accounts receivable	5	Tax	3
Cash holdings	5	Dividends	2
Short-term debt	20		
Accounts payable	25		



Current Assets = 60

Current Liabilities = 45

Current Ratio = $60/45 = 1.33$

Exercise – solution (2/3)

Compute and interpret (1) current ratio; (2) quick ratio, and (3) working capital to sales ratio

B/S		Income statement	
Owners' funds	40	Revenues	100
Fixed assets	160	Operating costs	75
Inventories	50	Depreciation	10
Long-term debt	135	Interest	2
Accounts receivable	5	Tax	3
Cash holdings	5	Dividends	2
Short-term debt	20		
Accounts payable	25		



Current Assets = 60

Current Liabilities = 45

Quick Ratio = $(60 - 50) / 45 = 0.22$

Exercise – solution (2/3)

Compute and interpret (1) current ratio; (2) quick ratio, and (3) working capital to sales ratio

B/S		Income statement	
Owners' funds	40	Revenues	100
Fixed assets	160	Operating costs	75
Inventories	50	Depreciation	10
Long-term debt	135	Interest	2
Accounts receivable	5	Tax	3
Cash holdings	5	Dividends	2
Short-term debt	20		
Accounts payable	25		



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Current Assets = 60

Current Liabilities = 45

Sales = 100

Working
capital to
sales ratio = $\frac{60 - 45}{100} * 100 = 15\%$

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FINANCIAL STRENGTH RATIOS



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Financial strength

- The company's ability to meet interest and principal payments in the long-run
- Financial strength ratios provide measures of this ability
- They are associated to the structure of capital:
 - risk capital (equity)
 - loan capital (bank and corporate debt)



Interest coverage ratio

$$\frac{\text{EBIT}}{\text{Interests}}$$

- It relates financial charges and profitability
 - how many times a firm's operating income exceeds interest expenses
- Three factors determine its value:
 - the operating profit,
 - the total amount borrowed
 - the effective rate of interest



The lower the ratio, the more the company is burdened by debt expense

Interest coverage ratio

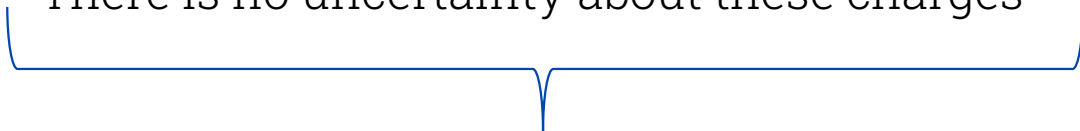
$$\frac{\text{EBIT}}{\text{Interests}}$$

- Can the ratio be negative?
 - What does it mean?
- <1 means bad financial situation
 - Why?
- <1.5, the ability to meet interest expenses is questionable
 - Worst situation: EBIT declines and interest increases over time



Structure of capital

- Composition of debt and equity
 - Where can we find the relevant items?
- What are the financial charges generated by debt?
 - Interest
 - Payment of the principal
 - There is no uncertainty about these charges



Certain cash-out flows vs uncertain
cash-in flows in a specific period



FINANCIAL
RISK



Structure of capital

- High corporate risk means higher likelihood of going out of business
 - If the firm is unable to meet debt payment: default
 - A court may impose supervision on how to handle a default: bankruptcy (i.e. legal procedure to pay creditors)
- But why then do companies use debt?
 - Couldn't firms just use equity (which is risky capital, i.e. if the firm doesn't make profit, shareholders don't get anything and cannot complain)
- Debt is usually cheaper than equity
 - Less risky for investors, expected return must be lower than equity, and average realized return must be lower too (i.e. interest rate paid to the lenders must be less than the rate of return earned by the shareholders)
 - Tax deductible
 - Good for funding growth



Debt to equity ratio (or LEVERAGE)

$$\frac{\text{Total liabilities} - \text{Owners' funds}}{\text{Owners' funds}}$$

- The entire debt is considered here, i.e. total liabilities minus owners' funds
- Values between 1 and 2 are typically good
- <1 indicate that the company is over-capitalized
- >2 may indicate excessive risk in the capital structure (high values common in capital intensive industries)



PROFITABILITY RATIOS

ROTA

$$\frac{\text{EBIT}}{\text{Total assets}} \%$$

ROS

$$\frac{\text{EBIT}}{\text{Total sales}} \%$$

Asset turnover

$$\frac{\text{Sales}}{\text{Total assets}}$$

ROE

$$\frac{\text{Net income}}{\text{Owners' funds}} \%$$

FINANCIAL STRENGTH RATIOS

Interest coverage

$$\frac{\text{EBIT}}{\text{Interests}}$$

D/E ratio

$$\frac{\text{Debt}}{\text{Owners' funds}}$$

Cost of interest

$$\frac{\text{Total interest}}{\text{Debt}} \%$$

LIQUIDITY RATIOS

Current ratio

$$\frac{\text{Current assets}}{\text{Current liabilities}}$$

Quick ratio

$$\frac{\text{Current assets} - \text{Inv.}}{\text{Current liabilities}}$$

Working capital/sales

$$\frac{\text{Current (Ass. - Liab.)}}{\text{Sales}} \%$$



5.0

FINANCIAL LEVERAGE



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Effect of financial leverage

$$\text{ROE} = \text{Net income/owners' funds}$$

— It tells us the value generated for shareholders

— We decompose ROE as:

$$(\text{Net income/sales}) \times (\text{sales/total assets}) \times (\text{total assets/equity})$$



ROS
(after tax and interest)



Asset turnover



Financial leverage
 $= (1 + \text{debt/equity})$



Effect of financial leverage

- An increase in debt/equity has two effects:
 - It increases ROE
 - It increases corporate risk
- It pays off to increase ROE by increasing debt/equity in good times
- But its very risky and not convenient to do so in bad times



Example

Current capital structure

Total Assets = \$1 million; Equity = \$1 million (10,000 shares);

Debt = \$0

New capital structure

Total Assets = \$1 million; Equity = \$700,000 (7,000 shares);

Debt = \$300,000

EBIT next year will be either \$50,000, \$125,000, or \$200,000 depending on macroeconomic conditions. The interest rate on debt is 10%. Ignore taxes

— How would this new capital structure affect the firm's ROE?



All equity	BAD	EXPECTED	GOOD
Debt	0	0	0
Equity	1 million	1 million	1 million
EBIT	50.000	125.000	200.000
Interest	0	0	0
Net income	50.000	125.000	200.000
ROE	5%	12.5%	20%

Recapitalization	BAD	EXPECTED	GOOD
Debt	300.000	300.000	300.000
Equity	700.000	700.000	700.000
EBIT	50.000	125.000	200.000
Interest	30.000	30.000	30.000
Net income	20.000	95.000	170.000
ROE	2.86%	13.57%	24.29%



Takeaway

- Use financial leverage with caution
 - A high use of financial leverage to deliver high ROE may also entail greater corporate risk
 - Particularly risky in times of high uncertainty
- Combine information from ROE with other indexes
 - In particular, ROTA, which assesses the ability to generate profits from the entire asset base – not just equity)
 - Also, use solvency indicators to better understand the capital structure of the company (too much debt?)



In class exercise

Calculate and comment on:

- ROTA, ROS and Asset turnover
- Current ratio and Quick ratio
- Working capital
- Debtor days, Creditor days, Inventory days
- Debt to equity ratio
- Interest coverage ratio
- ROE

Issued common stock	20
Net fixed assets	160
Operating costs	75
Inventories	50
Depreciation	10
Long-term debt	135
Accounts receivable	5
Cash	5
Capital reserve	10
Revenues	100
Short-term debt	20
Tax	2
Accounts payable	25
Dividends	1
Interest	11
Pre-paid insurance	20
Due dividends	20
Revenue reserve	10



THANKS.



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